MCBOMs Methodology

Mathematical Formulation and Validation

Texas A&M Transportation Institute

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Introduction

This document presents the mathematical formulation and validation of the Multidisciplinary Cost-Benefit Optimization Models (MCBOMs) framework, a Mixed-Integer Linear Programming (MILP) approach to transportation infrastructure investment decisions. Given a portfolio of candidate projects (sites, intersections, or corridors), each with multiple possible improvement alternatives, MCBOMs selects the combination of alternatives that maximizes total societal benefit subject to a budget constraint. The benefits include safety improvements (reductions in expected crash costs), operational improvements (reductions in travel time and vehicle operating costs), and corridor condition improvements (energy, environmental, accessibility, and resilience measures).

The framework is developed under FHWA Contract HRSO20240009PR by the Texas A&M Transportation Institute. It builds directly on Harwood, Rabbani, and Richard (2003), who developed the Resurfacing Safety Resource Allocation Program (RSRAP) for allocating resources to safety improvements within Resurfacing, Restoration, and Rehabilitation (RRR) projects on nonfreeway facilities, and on Banihashemi (2007), who extended that framework with intersection-level optimization using the IHSDM Crash Prediction Module. Default unit costs and economic parameters come from USDOT Benefit-Cost Analysis Guidance for Discretionary Grant Programs (May 2025).

This document is organized as follows. Section 2.1 establishes the foundational framework, including the project–alternative structure, the benefit definition, and the discounting convention. Section 2.2 presents the MILP formulation at the project and network levels. Sections 2.3 through 2.5 derive the safety, operational, and corridor condition benefit components. Section 2.6 documents the solution methodology, and Section 2.7 presents validation against the Harwood (2003) and Banihashemi (2007) benchmarks together with cross-validation against an internal benefit-cost analysis spreadsheet.

2.1 Foundational Framework

The MCBOMs framework addresses the multi-objective resource allocation problem faced by transportation agencies: given a portfolio of candidate projects, each with multiple possible improvement alternatives, determine the combination of alternatives that maximizes total societal benefit subject to a budget constraint. Three foundational design choices govern the formulation: the project–alternative structure, the benefit definition and quantification structure, and the discounting and time-horizon convention.

2.1.1 Project–Alternative Structure

A *project* represents a candidate location for investment. Examples include a corridor segment, an intersection, a bridge, or any other discrete asset for which an improvement decision can be made. Each project is denoted by the index $i \in \{1, 2, \ldots, I\}$, where I is the total number of projects in the analysis.

For each project i, the analyst defines a set of *improvement alternatives*, denoted $A_i = \{0, 1, 2, \ldots, J_i\}$. The alternative $j = 0$ is always the do-nothing baseline (no improvement im-

plemented at site i). Alternatives $j \geq 1$ each represent a distinct combination of treatments. Treatments may include, for example, lane widening, shoulder paving, intersection signalization, or pavement resurfacing. A single alternative may bundle multiple treatments, in which case the alternative's combined effect is computed using the appropriate combination rule for each benefit category (see Section 2.3 for safety, Section 2.4 for operations).

Each project is characterized by a record:

$$P_i = \{\text{Location}_i,\ \text{Cost}_{ij},\ \text{SafetyData}_{ij},\ \text{OperationData}_{ij},\ \text{CCMData}_{ij},\ \text{TreatmentType}_{ij}\} \tag{2.1}$$

for each alternative $j \in A_i$, capturing the data required to evaluate that alternative.

The optimization decision variable is binary:

$$x_{ij} \in \{0, 1\}, \qquad x_{ij} = \begin{cases} 1 & \text{if alternative } j \text{ is selected for project } i \\ 0 & \text{otherwise} \end{cases} \tag{2.2}$$

The mutual-exclusivity constraint $\sum_{j \in A_i} x_{ij} \leq 1$ ensures that at most one alternative (including do-nothing) is selected for each project.

2.1.2 Benefit Definition and Quantification Structure

The total present-value societal benefit of selecting alternative j for project i, evaluated over the analysis horizon T, is the discounted sum of three additive annual benefit components:

$$B_{ij}^{\text{total}} \;=\; \sum_{t=1}^{T} \left[\, B_{ijt}^{\text{safety}} + B_{ijt}^{\text{operations}} + B_{ijt}^{\text{corridor}} \,\right] \cdot DF_t \tag{2.3}$$

where:

- B_{ijt}^{safety} is the annual monetized safety benefit in year t (crash cost reduction); see Section 2.3.
- $B_{ijt}^{\text{operations}}$ is the annual monetized operational benefit in year t (travel time, vehicle operating costs, reliability); see Section 2.4.
- $B_{ijt}^{\text{corridor}}$ is the annual monetized corridor condition benefit in year t (energy, environmental, resilience, accessibility); see Section 2.5.
- $DF_t = 1/(1+r)^t$ is the discount factor in year t, with r the real discount rate (Section 2.1.4).

For notational compactness in the optimization formulation (Section 2.2), the present-value-aggregated forms are denoted B_{ij}^{safety}, $B_{ij}^{\text{operations}}$, and B_{ij}^{ccm}, with the relationship:

$$B_{ij}^{\text{total}} \;=\; B_{ij}^{\text{safety}} + B_{ij}^{\text{operations}} + B_{ij}^{\text{ccm}}$$

The additive decomposition assumes that the three benefit categories are independent of one another. Where overlap exists (for example, when a composite metric such as the

Mobility Energy Productivity (MEP) index embeds both travel time and energy components), explicit double-counting prevention rules are applied; see Section 2.5 for the relevant adjustments.

Each project–alternative pair has an associated cost:

$$C_{ij} = C_{ij}^{\text{construction}} + C_{ij}^{\text{indirect}} \tag{2.4}$$

where construction cost includes resurfacing and geometric improvements as applicable, and indirect cost captures traffic management during construction, environmental mitigation, and utility relocation. For the Task 4 prototype, indirect cost is folded into construction cost where data permits; otherwise it is set to zero.

Section 2.1.3 extends this cost definition to the segment level, showing how C_{ij} is composed from per-treatment per-segment cost contributions.

2.1.3 Alternative Composition and Segment-Level Treatment Specification

A central modeling concept in the MCBOMs framework, consistent with the segment-level structure used by Banihashemi (2007), is that an alternative j for project i is not necessarily a single uniform treatment applied across the entire project. Rather, an alternative is a *composition* of segment-specific treatments: different homogeneous segments within the same project may receive different treatment combinations under the same alternative.

Homogeneous segmentation. Each project i is partitioned into K_i homogeneous segments, where a segment $k \in \{1, 2, \ldots, K_i\}$ is a length of facility over which the relevant geometric, traffic, and crash-history attributes are approximately constant. New segments begin where the AADT, lane configuration, shoulder treatment, horizontal/vertical alignment, or roadside condition changes materially. This segmentation rule matches the Highway Safety Manual (AASHTO, 2010, Part C) homogeneous-segment definition and the construction Banihashemi (2007, page 112) uses for his Crash Prediction Module.

Treatment sets. For each alternative $j \in A_i$ and each segment $k \in \{1, \ldots, K_i\}$, the analyst defines a treatment set:

$$T_{jk} \;=\; \{\,\text{treatments applied to segment } k \text{ under alternative } j\,\}$$

For the do-nothing alternative ($j = 0$), $T_{0k} = \emptyset$ for every segment k. For non-trivial alternatives, T_{jk} may differ across segments within the same alternative; this is the segment-level composition property.

Alternative as a composition. An alternative j for project i is fully specified by the collection of its segment-level treatment sets:

$$\text{Alternative } j \text{ for project } i \;=\; \{\,T_{j1},\; T_{j2},\; \ldots,\; T_{jK_i}\,\}$$

Cost decomposition. The total cost of alternative j for project i is the sum of treatment-level costs across all segments:

$$C_{ij} = \sum_{k=1}^{K_i} \sum_{n \in T_{jk}} Cost_{ikn} \tag{2.5}$$

where $Cost_{ikn}$ is the implementation cost of treatment n on segment k of project i.

Benefit decomposition. Total present-value benefit B_{ij}^{total}, defined in Equation 2.3, similarly aggregates segment-level contributions. Each annual benefit component (B_{ijt}^{safety}, $B_{ijt}^{\text{operations}}$, $B_{ijt}^{\text{corridor}}$) is computed as a sum over segments $k = 1, \ldots, K_i$, with the segment-level value depending on the treatment set T_{jk} applied to that segment. The detailed segment-level expansions are given in Section 2.3 (Equation 2.18) for safety, Section 2.4 (Equation 2.23) for operations, and Section 2.5 (Equation 2.24) for corridor condition.

Optimization decision variable remains alternative-level. Although costs and benefits are computed at the treatment-segment level, the optimization decision variable x_{ij} remains binary at the alternative level. The optimizer selects an entire alternative (with its full collection of segment-specific treatments) or rejects it; partial alternatives are not selectable. This matches Harwood (2003)'s site-level decision-variable structure and is consistent with the Banihashemi (2007) formulation interpreted at the alternative level.

Illustrative example. Consider a 6-mile rural two-lane highway project ($i = 1$) partitioned into three 2-mile homogeneous segments ($K_1 = 3$). A single non-trivial alternative ($j = 1$) is defined as the following composition:

- Segment $k = 1$ (miles 0–2): left-turn lane addition at the intersection located within this segment. $T_{11} = \{\,\text{left-turn lane}\,\}$.
- Segment $k = 2$ (miles 2–4): centerline rumble strips. $T_{12} = \{\,\text{rumble strips}\,\}$.
- Segment $k = 3$ (miles 4–6): traffic signal installation at the intersection located within this segment. $T_{13} = \{\,\text{traffic signal}\,\}$.

The total cost of this alternative is computed by Equation 2.5:

$$\begin{aligned} C_{11} &= Cost_{1,1,\text{LTL}} + Cost_{1,2,\text{rumble}} + Cost_{1,3,\text{signal}} \\ &= \$300{,}000 + \$50{,}000 \cdot 2 \text{ mi} + \$250{,}000 \\ &= \$650{,}000 \end{aligned}$$

and total present-value benefit B_{11}^{total} is the sum of segment-level safety, operational, and corridor benefits computed under each segment's treatment set, discounted via Equation 2.3.

Combined-treatment effects within a segment. When a single segment k receives multiple treatments simultaneously ($|T_{jk}| > 1$), the combined safety effect uses the multiplicative Crash Modification Factor rule (Equation 2.20). For example, a segment receiving both shoulder widening (CMF = 0.93) and rumble strips (CMF = 0.86) has a combined CMF of $0.93 \times 0.86 = 0.80$. This rule is the HSM-recommended default; ongoing CMF Clearinghouse research is refining combined-treatment estimates.

Implementation note. The Task 4 prototype dataset encodes alternative composition through the alternative identifier itself (e.g., "Pavement Marking, Cable Barrier" as a single alternative label). A treatment-level cost catalog is required to evaluate Equation 2.5 term-by-term; in the prototype's BCA spreadsheet, this catalog is partially captured through per-mile unit cost fields by treatment type. Full treatment-level decomposition (in which each $Cost_{ikn}$ is computed from a separate treatment-cost table joined to a segment-treatment assignment table) is documented as a planned enhancement; the prototype currently uses alternative-level aggregate costs as an equivalent simplification.

2.1.4 Discounting and Analysis Horizon

All benefit and cost streams are discounted to present value using the USDOT-recommended real discount rate of 7 percent per year (USDOT, 2025, Section 4.3). The discount factor for year t is:

$$DF_t = \frac{1}{(1+r)^t}, \qquad r = 0.07 \tag{2.6}$$

For benefit streams that are constant across years (the standard assumption for the Task 4 prototype), the present-value calculation reduces to multiplication by the uniform-series present worth factor:

$$PWF(r, T) = \sum_{t=1}^{T} DF_t = \frac{(1+r)^T - 1}{r \cdot (1+r)^T} \tag{2.7}$$

At the framework defaults of $r = 0.07$ and $T = 20$ years, this evaluates to $PWF(0.07, 20) = 10.594$.

Analysis horizon. The Task 4 prototype uses a uniform 20-year analysis horizon for all alternatives, consistent with USDOT BCA Guidance (May 2025, Section 4.4) for capacity and operating improvements and with Banihashemi (2007), who applied a uniform 20-year project lifetime to all improvement alternatives in his case study.

Treatment-specific service life. Table 2.1 documents typical service lives by treatment type for reference. These values are not currently applied to the prototype's benefit calculations; all benefits are evaluated over the uniform 20-year horizon. This simplification applies uniformly across alternatives, which preserves the relative ranking that drives the optimization. Per-treatment service life and residual-value adjustments (in the sense of USDOT BCA Guidance Section 6.3, which applies residual value to construction costs rather than to benefit streams) are documented as planned enhancements.

Table 2.1: Reference service lives by treatment type. These values are documented for reference and for future model enhancements; they are not currently applied to the Task 4 prototype's benefit calculations, which use a uniform 20-year analysis horizon.

Treatment Type	**Reference Service Life (years)**
Pavement resurfacing	10
Lane widening	20
Shoulder widening / paving	20
Horizontal curve improvement	20
Vertical alignment improvement	20
Intersection signalization	15
Roundabout	25
Turn lane addition	20
Climbing/passing lane	20
Sight distance improvement	20

2.2 Mixed-Integer Linear Programming Formulation

The MCBOMs optimization problem is formulated as a Mixed-Integer Linear Program. This section presents the project-level formulation, in which only a single budget constraint applies, followed by the network-level formulation, which adds facility-type and regional sub-budget constraints.

2.2.1 Project-Level Formulation

The project-level MILP maximizes total net benefit subject to a single program-wide budget. The formulation consists of an objective function and three constraint families.

Objective function.

$$\max\ Z \;=\; \sum_{i=1}^{I} \sum_{j \in A_i} \left(B_{ij}^{\text{total}} - C_{ij}^{\text{disc}} \right) \cdot x_{ij} \tag{2.8}$$

The objective expresses total net societal benefit as the sum, over all projects and selected alternatives, of benefit minus the *discretionary* cost component C_{ij}^{disc}. The parentheses around $(B_{ij}^{\text{total}} - C_{ij}^{\text{disc}})$ are essential: the multiplication by x_{ij} applies to the entire net-benefit term, not to the cost alone.

Discretionary versus committed cost. For projects where project cost decomposes into a committed component C_{ij}^{comm} (expenditure that will occur regardless of the optimization outcome, typically because policy or asset condition mandates it) and a discretionary component C_{ij}^{disc} (expenditure subject to optimization choice), the objective subtracts only the

discretionary portion. The total cost remains $C_{ij} = C_{ij}^{\text{comm}} + C_{ij}^{\text{disc}}$ and is the quantity used in the budget constraint (Equation 2.9).

This distinction is consistent with Harwood et al. (2003), who model RRR projects in which pavement resurfacing is committed (every site is resurfaced under the program) and only the safety improvement cost is optimized. In that case:

$$\begin{aligned} C_{ij}^{\text{comm}} &= \text{resurfacing cost (committed by the RRR program)} \\ C_{ij}^{\text{disc}} &= \text{safety improvement cost (the optimization variable)} \end{aligned}$$

For projects where no committed component exists — $C_{ij}^{\text{comm}} = 0$ and $C_{ij}^{\text{disc}} = C_{ij}$ — Equation 2.8 reduces to the simple form $\max \sum (B_{ij}^{\text{total}} - C_{ij}) x_{ij}$, which is the case for the Banihashemi (2007) validation and for general projects without a fixed infrastructure baseline.

The MCBOMs implementation supports both conventions: the optimizer accepts a per-alternative objective coefficient, which the analyst sets to $(B_{ij}^{\text{total}} - C_{ij}^{\text{disc}})$ when modeling a Harwood-style problem and to $(B_{ij}^{\text{total}} - C_{ij})$ for a generic problem.

Budget constraint.

$$\sum_{i=1}^{I} \sum_{j \in A_i} C_{ij} \cdot x_{ij} \;\leq\; B^{\text{total}} \tag{2.9}$$

where B^{total} is the total program budget.

Mutual exclusivity.

$$\sum_{j \in A_i} x_{ij} \;\leq\; 1, \qquad \forall i \in \{1, 2, \ldots, I\} \tag{2.10}$$

This constraint allows at most one alternative (including do-nothing) to be selected for each project. When the inequality is strict ($\sum_j x_{ij} = 0$), the optimizer has implicitly chosen do-nothing for project i.

Binary decision variables.

$$x_{ij} \in \{0, 1\}, \qquad \forall i,\ \forall j \in A_i \tag{2.11}$$

Optional constraints. Beyond the four core constraints above, the project-level formulation supports three optional constraint families that reflect common real-world policy and engineering requirements. These constraints are activated only when the corresponding data is provided.

(a) Dependency constraints. If alternative j of project i requires the prior completion of alternative j' of project i' (for example, a corridor signalization plan that depends on an upstream intersection upgrade), the dependency is enforced by:

$$x_{ij} \;\leq\; x_{i'j'} \tag{2.12}$$

(b) Cross-project exclusivity. If alternatives at different projects conflict (for example, when two adjacent projects compete for the same right-of-way or land use), the conflict is enforced by:

$$x_{ij} + x_{i'j'} \;\leq\; 1 \tag{2.13}$$

(c) Minimum benefit-cost ratio. To require selected alternatives to exceed a minimum benefit-cost ratio:

$$\sum_{j \in A_i} B_{ij}^{\text{total}} \cdot x_{ij} \;\geq\; \theta \cdot \sum_{j \in A_i} C_{ij} \cdot x_{ij}, \qquad \forall i \tag{2.14}$$

where θ is the minimum acceptable benefit-cost ratio (for example, $\theta = 1.0$ enforces positive net benefit at every selected project; $\theta = 2.0$ enforces a 2:1 benefit-cost ratio).

Implementation status of optional constraints. The MCBOMs prototype optimizer implements all four core constraints (Equations 2.8–2.11) and all three optional project-level constraints (Equations 2.12, 2.13, and 2.14). The optimizer also implements the network-level constraints (Equations 2.15–2.17). Optional constraints are activated only when the corresponding agency data is provided; absent that data, they remain inactive and impose no restriction. Notebook 03 in the project repository (`notebooks/03_optimizer_capabilities.ipynb`) demonstrates each optional and network-level constraint operating on a synthetic six-site test network. The Harwood (2003) and Banihashemi (2007) validation case studies do not exercise dependency or cross-project exclusivity constraints; minimum-BCR enforcement is optional and disabled by default.

Facility-type scope. The project-level optimization framework applies to the following facility types, with facility-specific Safety Performance Functions (SPFs), Crash Modification Factors (CMFs), and operational performance parameters as defined in Section 2.3 (safety) and Section 2.4 (operations):

- **Rural highways:** two-lane highways and intersections (including roundabouts); four-lane divided and undivided highways and intersections; freeways, ramps, and ramp terminals.
- **Urban / suburban facilities:** two-lane highways and intersections; multilane highways and intersections; one-way arterials and intersections; freeways, ramps, and ramp terminals.

2.2.2 Network-Level Formulation

The network-level formulation extends the project-level MILP by adding facility-type sub-budget constraints, regional cap constraints, and regional minimum-investment constraints. These additional constraints reflect common real-world resource allocation requirements, such as ensuring that no single facility class consumes a disproportionate share of the budget, or that each geographic region receives a minimum allocation of investment.

Sets for network-level constraints. Let $\mathcal{F}_m$ denote the set of project indices belonging to facility type $m \in \{m_1, m_2, \ldots, m_M\}$ (e.g., rural two-lane, rural multilane, urban arterial). Let $\mathcal{R}_r$ denote the set of project indices located in region $r \in \{r_1, r_2, \ldots, r_R\}$ (e.g., District 1, District 2).

Objective function. Identical to the project-level objective (Equation 2.8); the summation index i now ranges over all projects across all facility types and regions.

Total budget constraint. Identical to Equation 2.9.

Mutual exclusivity. Identical to Equation 2.10.

Facility-type sub-budget constraints.

$$\sum_{i \in \mathcal{F}_m} \sum_{j \in A_i} C_{ij} \cdot x_{ij} \;\leq\; B_m, \qquad \forall m \tag{2.15}$$

where B_m is the maximum allocation to facility type m.

Regional sub-budget constraints (cap).

$$\sum_{i \in \mathcal{R}_r} \sum_{j \in A_i} C_{ij} \cdot x_{ij} \;\leq\; B_r, \qquad \forall r \tag{2.16}$$

Regional minimum-investment constraints (floor).

$$\sum_{i \in \mathcal{R}_r} \sum_{j \in A_i} C_{ij} \cdot x_{ij} \;\geq\; \beta_r \cdot B^{\text{total}}, \qquad \forall r \tag{2.17}$$

where $\beta_r \in [0, 1]$ is the minimum-investment fraction for region r. Setting $\beta_r = 0$ disables the floor constraint for that region.

Binary decision variables. Identical to Equation 2.11.

2.2.3 Worked Example: Project-Level Optimization for Three Rural Highway Segments

To illustrate the project-level MILP, consider three rural two-lane highway projects with the following alternative inventory and a program budget of \$2,000,000. Costs and total present-value benefits are expressed in thousands of dollars.

Project 1 (5-mile rural segment):

- $j = 0$ (do nothing): $C_{10} = 0$, $B_{10}^{\text{total}} = 0$
- $j = 1$ (widen lanes to 12 ft): $C_{11} = \$800$, $B_{11}^{\text{total}} = \$1{,}200$

- $j = 2$ (widen shoulders to 6 ft): $C_{12} = \$600$, $B_{12}^{\text{total}} = \$950$

Project 2 (3-mile rural segment with curve):

- $j = 0$: $C_{20} = 0$, $B_{20}^{\text{total}} = 0$
- $j = 1$ (flatten curve): $C_{21} = \$1{,}500$, $B_{21}^{\text{total}} = \$2{,}400$

Project 3 (2-mile rural segment):

- $j = 0$: $C_{30} = 0$, $B_{30}^{\text{total}} = 0$
- $j = 1$ (rumble strips): $C_{31} = \$100$, $B_{31}^{\text{total}} = \$150$
- $j = 2$ (widen lanes + shoulders): $C_{32} = \$1{,}200$, $B_{32}^{\text{total}} = \$1{,}800$

MILP instance. The objective function and constraints specialize to:

$$\begin{aligned}
\text{Maximize:} \quad & Z = 400x_{11} + 350x_{12} + 900x_{21} + 50x_{31} + 600x_{32} \\
\text{Subject to:} \quad & 800x_{11} + 600x_{12} + 1500x_{21} + 100x_{31} + 1200x_{32} \;\leq\; 2000 \\
& x_{10} + x_{11} + x_{12} \;\leq\; 1 \\
& x_{20} + x_{21} \;\leq\; 1 \\
& x_{30} + x_{31} + x_{32} \;\leq\; 1 \\
& x_{ij} \in \{0, 1\}
\end{aligned}$$

where the objective coefficients are net benefits ($B_{ij}^{\text{total}} - C_{ij}$) for non-zero alternatives. This worked example has no committed cost component ($C_{ij}^{\text{comm}} = 0$, $C_{ij}^{\text{disc}} = C_{ij}$), so the objective uses the simple form.

Optimal solution. A MILP solver evaluates all feasible combinations of project–alternative selections and returns the combination that maximizes total net benefit subject to the budget constraint. The optimal solution for this instance is:

- $x_{11} = 1$ (Project 1, lane widening) – net benefit \$400, cost \$800
- $x_{20} = 1$ (Project 2, do nothing) – net benefit \$0, cost \$0
- $x_{32} = 1$ (Project 3, lane + shoulder widening) – net benefit \$600, cost \$1,200

Total cost: $\$800 + \$0 + \$1{,}200 = \$2{,}000$ (at budget). **Total net benefit:** $\$400 + \$600 = \$1{,}000$ (thousand dollars).

This example illustrates two key features of the MILP formulation: (a) the trade-off between high-cost / high-benefit alternatives (Project 2's curve flattening is rejected despite its \$900 net benefit because its \$1,500 cost would crowd out higher-net-benefit combinations elsewhere); and (b) the role of the do-nothing alternative in providing a feasible default when budget constraints prevent funding any improvement at a given site.

2.3 Safety Benefit Quantification

The safety benefit module quantifies the present-value reduction in expected crash costs resulting from each improvement alternative. The formulation follows the methodology of the Highway Safety Manual (AASHTO, 2010, Part C), extended to the optimization context developed by Harwood et al. (2003) and Banihashemi (2007), and updated to use current USDOT-recommended unit crash costs.

2.3.1 Severity Disaggregation

The MCBOMs framework adopts the five-bin KABCO crash severity classification as the standard:

- **K** — Killed (fatal injury)
- **A** — Suspected serious (incapacitating) injury
- **B** — Suspected minor (non-incapacitating) injury
- **C** — Possible injury
- **O** — Property damage only (no injury)

The severity index set is denoted $\mathcal{S} = \{K, A, B, C, O\}$. All benefit calculations sum over this five-element set.

Variants for benchmark reproduction. Two legacy severity conventions are used in the validation reproductions described in Section 2.7: Harwood (2003) used a two-bin convention (fatal+injury combined; PDO), and Banihashemi (2007) collapsed five severity levels into a single average crash cost per facility type. These variants are documented as conversion bridges and applied only when reproducing the respective papers' published results.

2.3.2 Unit Crash Costs

The framework default unit crash costs are taken from USDOT BCA Guidance (May 2025), Appendix A, Table A-1 (page 39), in 2023 dollars:

$$UC_K = \$13{,}200{,}000, \quad UC_A = \$1{,}254{,}700, \quad UC_B = \$246{,}900, \quad UC_C = \$118{,}000, \quad UC_O = \$5{,}300$$

These values include both economic costs (medical, vehicle damage, lost wages) and quality-adjusted-life-year (QALY) costs. State agencies applying the MCBOMs framework may substitute state-specific crash costs in place of the national defaults; FHWA-SA-25-021 (October 2025), Step 5, provides a per-capita-income adjustment procedure for converting national values to state-level values.

2.3.3 Safety Benefit Equation

The present-value safety benefit for project i, alternative j, over the analysis horizon T is:

$$B_{ij}^{\text{safety}} = \sum_{t=1}^{T} \sum_{k=1}^{K_i} \sum_{s \in \mathcal{S}} \left[E_{ikst}^{\text{no-build}} - E_{ijkst}^{\text{build}} \right] \cdot UC_s \cdot DF_t \tag{2.18}$$

where:

- K_i is the number of homogeneous segments (or intersections) composing project i, per HSM Part C segmentation
- $E_{ikst}^{\text{no-build}}$ is the expected number of crashes of severity s on segment k in year t under the no-build (existing) condition
- E_{ijkst}^{build} is the corresponding expected count under the build condition (alternative j implemented)
- UC_s is the unit crash cost for severity s
- DF_t is the discount factor from Equation 2.6

The build-condition expected crash count is obtained by applying the relevant Crash Modification Factor (CMF):

$$E_{ijkst}^{\text{build}} = E_{ikst}^{\text{no-build}} \cdot CMF_{ijks} \tag{2.19}$$

For alternatives that bundle multiple individual treatments $n = 1, 2, \ldots, N_{ij}$, the combined CMF is computed using the HSM-recommended multiplicative rule:

$$CMF_{ij}^{\text{combined}} = \prod_{n=1}^{N_{ij}} CMF_n \tag{2.20}$$

This rule assumes that individual treatment effects are independent. The HSM acknowledges this as a known simplification; ongoing research in the FHWA CMF Clearinghouse is refining combined CMF estimates, but no consensus alternative exists at the time of this writing.

2.3.4 Constant-Annual-Crashes Simplification

For the Task 4 prototype, the no-build expected crash count is assumed constant across years (no traffic growth or treatment-effectiveness decay):

$$E_{ikst}^{\text{no-build}} = E_{iks}^{\text{no-build}}, \qquad \forall t \in \{1, 2, \ldots, T\}$$

Under this assumption, Equation 2.18 factors as:

$$B_{ij}^{\text{safety}} = \left[\sum_{k=1}^{K_i} \sum_{s \in \mathcal{S}} \left(E_{iks}^{\text{no-build}} - E_{ijks}^{\text{build}} \right) \cdot UC_s \right] \cdot PWF(r, T) \tag{2.21}$$

This is the form implemented in the MCBOMs Python module `mcboms.benefits.safety`. Time-varying expectations (for traffic growth or effectiveness decay) are supported by an alternative function (`safety_benefit_time_varying`) but are not exercised in the Task 4 prototype.

2.3.5 Source of $E^{\text{no-build}}$ for the Prototype

For the Task 4 prototype, the no-build expected crash count $E_{iks}^{\text{no-build}}$ is drawn directly from observed crash counts (typically a 3- to 5-year crash history annualized). This convention matches:

- Banihashemi (2007), who explicitly states: "This optimization model is for cases in which the EB procedure is not appropriate or there are not enough crash data to support this process" (page 6).
- Harwood (2003), who uses observed crash counts subdivided by default severity proportions (page 4).
- The parallel data-curation BCA spreadsheet developed in Task 4, which uses observed crashes directly.

Empirical Bayes adjustment of $E^{\text{no-build}}$ per Hauer (1997) is a documented methodological option for implementations with adequate crash history data. EB adjustment, when applied, operates as a preprocessing step that produces an EB-adjusted $E^{\text{no-build}}$ consumed unchanged by Equation 2.18. The MCBOMs framework neither requires nor forbids EB adjustment; it is a deployment-time choice.

2.3.6 Connection to the Optimization Decision Variable

The safety benefit B_{ij}^{safety} defined in Equation 2.18 is computed during preprocessing for every alternative j of every project i. The optimizer's objective function (Equation 2.8) multiplies B_{ij}^{safety} by the binary decision variable x_{ij}, ensuring that benefits are credited only when an alternative is actually selected.

The expected crash count that materializes after the optimization completes is given by the gating identity:

$$E_{iks}^{\text{final}} = \left(1 - \sum_{j \in A_i} x_{ij}\right) E_{iks}^{\text{no-build}} + \sum_{j \in A_i} x_{ij} \cdot E_{ijks}^{\text{build}} \tag{2.22}$$

with $\sum_{j \in A_i} x_{ij} \leq 1$. This identity is included for expository completeness; it is not needed in computation, since B_{ij}^{safety} is precomputed and the optimizer operates on Equation 2.8 directly.

2.3.7 Worked Example

To illustrate the application of Equation 2.21, consider a single rural two-lane segment with the following characteristics:

- Segment length: 5 miles
- AADT: 5,000 vehicles per day
- Observed annual crash frequency $E^{\text{no-build}} = 6.0$ crashes per year

- Severity distribution (from observed crash history): 1.3% K, 6.5% A, 9.5% B, 11.7% C, 71.0% O
- Proposed treatment: lane widening from 10 ft to 12 ft, with a single CMF of 0.88 applied uniformly across all severities
- Discount rate $r = 0.07$, analysis horizon $T = 20$ years

Step 1: Disaggregate observed crashes by severity.

$$E_K^{\text{no-build}} = 0.078, \quad E_A^{\text{no-build}} = 0.390, \quad E_B^{\text{no-build}} = 0.570, \quad E_C^{\text{no-build}} = 0.702, \quad E_O^{\text{no-build}} = 4.260$$

(crashes per year)

Step 2: Apply the CMF to obtain build-condition expected crashes. With $CMF = 0.88$:

$$E_K^{\text{build}} = 0.0686, \quad E_A^{\text{build}} = 0.3432, \quad E_B^{\text{build}} = 0.5016, \quad E_C^{\text{build}} = 0.6178, \quad E_O^{\text{build}} = 3.7488$$

Step 3: Compute annual benefit (sum over severity). The reduction is $(1 - CMF) = 0.12$ of each severity bin. Multiplying by the USDOT BCA May 2025 unit costs:

$$\begin{aligned}
\text{Annual safety benefit} &= (0.12)(0.078)(\$13{,}200{,}000) \\
&+ (0.12)(0.390)(\$1{,}254{,}700) \\
&+ (0.12)(0.570)(\$246{,}900) \\
&+ (0.12)(0.702)(\$118{,}000) \\
&+ (0.12)(4.260)(\$5{,}300) \\
&= \$123{,}552 + \$58{,}720 + \$16{,}888 + \$9{,}940 + \$2{,}709 \\
&= \$211{,}809 \text{ per year}
\end{aligned}$$

Step 4: Convert to present value.

$$B^{\text{safety}} = \$211{,}809 \times PWF(0.07, 20) = \$211{,}809 \times 10.594 = \$2{,}243{,}913$$

This worked example uses the framework default unit crash costs (USDOT BCA May 2025). The MCBOMs Python implementation reproduces this calculation exactly; see Section 2.7.1 for the validation result.

2.4 Operational Benefit Quantification

The operational benefit module quantifies the present-value reduction in travel time and vehicle operating costs resulting from each improvement alternative. The formulation follows USDOT BCA Guidance (May 2025, Section 5.2 and Appendix A Tables A-2 and A-3) and generalizes the operational-benefit treatments in Harwood (2003)'s PTOB term and Banihashemi (2007)'s delay-cost term, both of which are recovered as special cases of the MCBOMs formulation.

2.4.1 Operational Benefit Equation

The present-value operational benefit for project i, alternative j, is:

$$B_{ij}^{\text{operations}} = \sum_{t=1}^{T} \sum_{k=1}^{K_i} \left[\sum_{v \in \mathcal{V}} \Delta d_{ijkvt} \cdot AADT_{ijkvt} \cdot 365 \cdot OCC_v \cdot VOT_v + \sum_{v \in \mathcal{V}} \Delta VOC_{ijkv} \cdot VMT_{ijkvt} \right] \cdot DF_t \tag{2.23}$$

where:

- Δd_{ijkvt} is the change in per-vehicle delay (hours per vehicle), for vehicle class v on segment k in year t, attributable to alternative j at project i. Computed externally via Highway Capacity Manual procedures, IHSDM, TWOPAS, or microsimulation.
- $AADT_{ijkvt}$ is the annual average daily traffic by vehicle class (vehicles per day)
- 365 is the annualization factor (days per year)
- OCC_v is the average vehicle occupancy by vehicle class (persons per vehicle), drawn from USDOT BCA Guidance Appendix A Table A-3
- VOT_v is the value of travel time by vehicle class ($ per person-hour), drawn from USDOT BCA Guidance Appendix A Table A-2
- ΔVOC_{ijkv} is the change in vehicle operating cost ($ per vehicle-mile)
- VMT_{ijkvt} is the vehicle-miles traveled per year, by segment and vehicle class
- DF_t is the discount factor from Equation 2.6
- $\mathcal{V}$ is the set of vehicle classes (e.g., light-duty passenger, single-unit truck, combination truck)

Dimensional consistency. The travel time term has units (hr/veh) · (veh/day) · (days) · (persons/veh) · ($/person-hr) = $. The vehicle operating cost term has units ($/veh-mi) · (veh-mi) = $.

Standard parameter values. For the Task 4 prototype, the following USDOT BCA Guidance May 2025 values apply by default:

- VOT (all-purposes blend, passenger): $21.10 per person-hour (Table A-2, page 40)
- OCC (passenger vehicles, all travel): 1.52 persons per vehicle (Table A-3, page 41)
- VOC (light-duty vehicles): $0.56 per vehicle-mile (Table A-4, page 41)
- VOC (commercial trucks): $1.27 per vehicle-mile (Table A-4)

2.4.2 Generalization of Harwood and Banihashemi Operational Treatments

Equation 2.23 generalizes the operational-benefit treatments in the two benchmark studies; both are recovered as special cases.

Harwood (2003) PTOB as a special case. Harwood's Present-value Traffic Operational Benefit (PTOB) captures the travel time savings from a 1 mph speed increase observed after pavement resurfacing, valued at \$10.00 per hour (1994 dollars), assumed to persist for 30 months. Setting $OCC_v = 1$, $VOT_v = \$10.00/\text{hr}$, $\Delta VOC = 0$, and limiting the temporal sum to 30 months ($T = 2.5$ years) at Harwood's 4 percent discount rate recovers PTOB exactly.

Banihashemi (2007) delay cost as a special case. Banihashemi minimizes the sum of crash and delay costs, with delay valued at \$12.50 per vehicle-hour (page 109), computed as \$10/hr $\times$ 1.25 persons per vehicle (occupancy folded into a single rate). Delay times D are computed externally in vehicle-hours per year using IHSDM/TWOPAS for highway segments and Highway Capacity Manual procedures (Equations A30-11 and A30-15) for intersections. Setting $OCC_v \cdot VOT_v = \$12.50/\text{veh-hr}$ (a single combined rate), $\Delta VOC = 0$, and treating $\Delta d \cdot AADT \cdot 365$ as Banihashemi's annual D recovers his delay cost term exactly.

This generalization allows the MCBOMs framework to be applied across a wider range of project types than either benchmark addresses individually, while remaining consistent with both formulations as special cases.

2.4.3 Worked Example

To illustrate the application of Equation 2.23, consider a single highway segment with the following characteristics:

- Segment length: 5 miles
- AADT: 1,500 vehicles per day (passenger vehicles only)
- Per-vehicle delay reduction: $\Delta d = 17$ seconds per vehicle $= 17/3600 = 0.00472$ hr per vehicle
- Average occupancy $OCC = 1.52$ persons per vehicle
- Value of travel time $VOT = \$21.10$ per person-hour (USDOT BCA Guidance May 2025, all-purposes)
- Vehicle operating cost change $\Delta VOC = 0$ (no change in operating cost; the alternative reduces delay only)
- Discount rate $r = 0.07$, analysis horizon $T = 20$ years

Step 1: Compute annual person-hours of delay reduction.

$$\begin{aligned}\text{Annual person-hours saved} &= \Delta d \cdot AADT \cdot 365 \cdot OCC \\ &= 0.004722 \cdot 1{,}500 \cdot 365 \cdot 1.52 \\ &= 3{,}930 \text{ person-hours per year}\end{aligned}$$

Step 2: Monetize.

$$\text{Annual operational benefit} = 3{,}930 \times \$21.10 = \$82{,}919 \text{ per year}$$

Step 3: Convert to present value.

$$B^{\text{operations}} \;=\; \$82{,}919 \times PWF(0.07, 20) \;=\; \$82{,}919 \times 10.594 \;=\; \$878{,}449$$

This worked example uses the framework default parameters. The vehicle operating cost contribution is zero in this case; for projects that change vehicle operating cost (for example, by altering the route's grade or surface type), the second term in Equation 2.23 contributes additionally.

2.5 Corridor Condition Benefit Quantification

Corridor Condition Measures (CCMs) capture benefit categories that fall outside the conventional safety and operational scope: energy and emissions reductions, environmental and health co-benefits, resilience to natural hazards, accessibility improvements, and changes in property value. The MCBOMs framework provides a generic computational structure for CCM benefits and a reference table of monetization rates.

2.5.1 CCM Benefit Computation Structure

The present-value CCM benefit is computed as:

$$B_{ij}^{\text{ccm}} \;=\; \sum_{c \in \mathcal{C}} \sum_{t=1}^{T} \sum_{k=1}^{K_i} \Delta Q_{ijkt}^{c} \cdot V_t^c \cdot DF_t \tag{2.24}$$

where:

- $\mathcal{C}$ is the set of CCM categories applicable to the project
- ΔQ_{ijkt}^{c} is the change in the physical quantity for category c on segment k in year t attributable to alternative j at project i (e.g., metric tons of NO_x, gallons of fuel, expected damage from flood events)
- V_t^c is the monetization rate for category c in year t (\$ per unit of Q^c)
- DF_t is the discount factor from Equation 2.6

2.5.2 CCM Categories and Reference Sources

The Task 4 prototype identifies eight CCM categories drawn from the methodology selected in Task 3. For each, Table 2.2 identifies the physical quantity to be measured and the recommended monetization source.

Table 2.2: CCM categories: physical quantities and monetization sources.

Category	**Physical Quantity (ΔQ^c)**	**Monetization Source (V^c)**
Energy / Fuel	Annual fuel consumption change (gallons or BTU)	USDOT BCA Guidance Appendix A Table A-4 (vehicle operating cost separation; reflected in $B^{\text{operations}}$ when applicable)
Air Quality (NOx, SOx, $PM_{2.5}$)	Annual emissions change (metric tons)	USDOT BCA Guidance Appendix A Table A-6 (page 42)
Resilience to natural hazards	Expected annual damages avoided ($) or service-loss days reduced	FEMA Standard Economic Values v13 (September 2024)
Accessibility / Mobility Energy Productivity (MEP)	Composite mobility index change (per FDOT/NREL methodology)	Jin et al. (2023), FDOT BDV29-977-66
Reduced Emergency Service Delay	Avoided property damage from fire-service delay; ambulance survival rates	FEMA methodology adapted for transportation (USDOT BCA Guidance Section 5.7)
Stormwater Runoff	Reduced runoff volume (gallons) and pollutant loads	Local utility cost data; USDOT BCA Guidance Section 5.7
Wildlife Crash Reduction	Wildlife-vehicle collisions avoided	USDOT BCA Guidance Section 5.7 (treated within safety methodology where applicable)
Property Value Increase	Net land value change attributable to project ($, one-time)	Local real estate data; USDOT BCA Guidance Section 5.7

2.5.3 Double-Counting Prevention

The additive decomposition $B_{ij}^{\text{total}} = B_{ij}^{\text{safety}} + B_{ij}^{\text{operations}} + B_{ij}^{\text{ccm}}$ assumes that the three benefit categories are independent. Three mechanisms enforce this independence:

(1) Data-level tagging. Each benefit observation entering the calculation is tagged with a category indicator $\delta_{c,ikt} \in \{0, 1\}$ identifying its source category. An observation contributes to exactly one of B^{safety}, $B^{\text{operations}}$, or B^{ccm}.

(2) Equation-level gating. Each component sums only over its tagged observations. For example, fuel-consumption changes that are already reflected in the operational VOC term

of Equation 2.23 are not separately counted in the energy CCM category.

(3) Composite-metric adjustment. When a composite index is used that embeds multiple component categories (for example, the FDOT/NREL MEP index, which combines travel time, reliability, and energy with weights $w_{TT} = 0.40$, $w_{Rel} = 0.25$, $w_{Energy} = 0.35$), the operational and corridor components are reduced proportionally to avoid duplicating the embedded contributions:

$$\begin{aligned} B_{ij}^{\text{operations}} &= (1 - w_{TT}) \cdot B_{ij}^{TT} \;+\; (1 - w_{Rel}) \cdot B_{ij}^{Rel} \;+\; B_{ij}^{SD} \\ B_{ij}^{\text{ccm}} &= (1 - w_{Energy}) \cdot B_{ij}^{Energy} \;+\; B_{ij}^{\text{other-CCM}} \end{aligned}$$

Validation. The pairwise correlation between benefit components across all (i, j) pairs is monitored as a diagnostic check; correlations satisfying $|\rho(B^{c_1}, B^{c_2})| > 0.3$ are flagged for manual review of potential double-counting.

2.6 Solution Methodology

This section describes the solution approach for the MCBOMs MILP, the software implementation, and the relationship between the MILP formulation and alternative optimization paradigms considered during model selection.

2.6.1 Mixed-Integer Linear Programming Formulation Summary

The complete MCBOMs MILP is summarized below. The formulation has four core constraint families that are always active and three optional constraint families that are activated

only when the corresponding agency data is provided.

$$\text{Maximize:} \quad Z = \sum_{i=1}^{I} \sum_{j \in A_i} (B_{ij}^{\text{total}} - C_{ij}^{\text{disc}}) \cdot x_{ij} \qquad \text{(objective)}$$

$$\text{Subject to:} \quad \sum_{i=1}^{I} \sum_{j \in A_i} C_{ij} \cdot x_{ij} \leq B^{\text{total}} \qquad \text{(total budget)}$$

$$\sum_{j \in A_i} x_{ij} \leq 1, \qquad \forall i \qquad \text{(mutual exclusivity)}$$

$$x_{ij} \in \{0, 1\}, \qquad \forall i, \forall j \in A_i \qquad \text{(binary)}$$

$$\sum_{i \in \mathcal{F}_m} \sum_{j \in A_i} C_{ij} \cdot x_{ij} \leq B_m, \quad \forall m \qquad \text{(facility-type sub-budget)}$$

$$\sum_{i \in \mathcal{R}_r} \sum_{j \in A_i} C_{ij} \cdot x_{ij} \leq B_r, \quad \forall r \qquad \text{(regional cap)}$$

$$\sum_{i \in \mathcal{R}_r} \sum_{j \in A_i} C_{ij} \cdot x_{ij} \geq \beta_r B^{\text{total}}, \quad \forall r \qquad \text{(regional floor)}$$

$$x_{ij} \leq x_{i'j'} \qquad \text{(dependency, optional)}$$

$$x_{ij} + x_{i'j'} \leq 1 \qquad \text{(cross-project exclusivity, optional)}$$

$$\sum_{j \in A_i} B_{ij}^{\text{total}} \cdot x_{ij} \geq \theta \sum_{j \in A_i} C_{ij} \cdot x_{ij}, \quad \forall i \qquad \text{(min BCR, optional)}$$

The cost decomposition $C_{ij} = \sum_{k=1}^{K_i} \sum_{n \in T_{jk}} Cost_{ikn}$ (Equation 2.5) is implicit in the above; total benefit B_{ij}^{total} is computed per Equation 2.3 with components from Sections 2.3–2.5.

Problem scale. The MILP has $\sum_{i=1}^{I} |A_i|$ binary decision variables. The core-constraint count comprises one total-budget constraint, I mutual-exclusivity constraints, $|\mathcal{F}|$ facility-type sub-budgets, and up to $2|\mathcal{R}|$ regional constraints (cap and floor). The optional constraints add at most one inequality per declared dependency pair, conflict pair, or BCR-enforced project. For the Harwood (2003) ten-site validation case, the problem has 41 binary variables and 11 constraints; for the Banihashemi (2007) full network case (135 homogeneous segments plus 13 intersections), the published problem has 3,385 binary variables and 535 constraints.

2.6.2 Software Implementation

The MCBOMs MILP is implemented as a Python package, `mcboms`, that constructs the model from input data and solves it via Gurobi (commercial) or CBC (open-source) backends. The MILP itself — the mathematical statement summarized in Section 2.6.1 — is solver-agnostic; it can be exported to standard MILP exchange formats (AMPL `.mod`/`.dat`, GAMS `.gms`, CPLEX `.lp`, MPS) and solved by any compatible commercial or open-source MILP solver, including IBM CPLEX, FICO Xpress, Gurobi, SCIP, and CBC. Reference instances of the

Harwood and Banihashemi validation case studies are provided in these formats so that any reviewer can independently reproduce the optimization results without depending on the Python implementation.

The implementation is organized into a modular Python package, `mcboms`, with the following structure (status as of Task 4 prototype):

- `mcboms.core` — optimizer engine, alternative-set enumeration, and solver back-end interface (*implemented*).
- `mcboms.benefits.safety` — implementation of Equation 2.21 (*implemented*, validated against the worked example to within floating-point precision and cross-validated against the parallel BCA spreadsheet to $0.00 error across 35 segment–alternative pairs).
- `mcboms.benefits.operations` — implementation of Equation 2.23 (*implemented*; 26 unit tests in `tests/test_operations.py` covering the worked example, default unit values, dimensional consistency, and the DataFrame batch interface).
- `mcboms.benefits.ccm` — implementation of Equation 2.24 (*implemented*; 34 unit tests in `tests/test_ccm.py` covering each per-category monetization function and the double-counting prevention against the operational benefit module).
- `mcboms.io` — data I/O modules for HSM-format input, spreadsheet input, and validation case studies (*implemented* for the BCA spreadsheet reader; HSM-format ingestion to follow).
- `mcboms.utils.economics` — discount rates, present-worth factors, and unit crash cost defaults (USDOT BCA Guidance May 2025; *implemented*).
- `mcboms.data` — bundled validation case study data for Harwood (2003) and Banihashemi (2007) intersection sub-problem (*implemented*).

The full source code is available in the project's GitHub repository, along with an automated test suite (109 tests covering the worked example, Harwood reproduction, and per-component unit tests), validation scripts (`run_harwood_validation.py`, `banihashemi_intersections.py`) and a programmer's guide.

2.6.3 Comparative Summary of Solution Methodologies

Task 3 evaluated four candidate solution methodologies for the MCBOMs problem: MILP, surrogate-assisted genetic algorithms (GA), reinforcement learning, and machine-learning regression on pre-computed benefit surfaces. MILP was selected as the production methodology on the following bases:

- **Global optimality.** MILP solvers provide certified globally optimal solutions (within numerical tolerance), which is required for FHWA-deliverable resource allocation results.
- **Methodological transparency.** The objective function and constraints are explicit and auditable. Each step in the optimization can be traced back to the formulation, in contrast to black-box machine learning approaches.

- **Compatibility with HSM and CMF methodology.** The MILP formulation operates on benefit and cost values computed from HSM-standard methods (SPFs, CMFs, EB adjustment when applied), preserving the methodological lineage that FHWA deliverables require.
- **Maturity.** MILP for transportation resource allocation has a 30-year track record (Cohon, 1978; Harwood, 2003; Banihashemi, 2007; AASHTO Safety Analyst). Solver technology is mature.
- **Tractability.** The Harwood (2003) and Banihashemi (2007) case studies solve in under one second on commodity hardware. Realistic state-DOT problems with 10^4–10^5 variables solve in minutes.

Machine learning methods are explicitly retained as preprocessing tools where helpful (for example, gap-filling missing inputs to HSM SPFs), but are not used as substitutes for the optimization core.

2.7 Validation Against Published Benchmarks

The MCBOMs implementation has been validated at three independent levels: a methodology-level reproduction of the worked example (Section 2.3.7), a program-level reproduction of Harwood et al. (2003), and a network-level reproduction of Banihashemi (2007). Each validation level exercises a distinct portion of the framework and tests against external published benchmarks rather than internal self-consistency.

2.7.1 Methodology-Level Validation: Worked Example

The worked example in Section 2.3.7 computes a single-segment safety benefit by direct application of Equation 2.21. The MCBOMs Python implementation (`mcboms.benefits.safety.safety_benef` reproduces this calculation exactly: the annual safety benefit is computed as $211,809, and the 20-year present-value benefit is $2,243,913. The implementation matches the manual calculation to within numerical floating-point precision (maximum discrepancy less than $0.01 across all severity components).

This validation confirms that the safety benefit module correctly implements the methodology described in this document and that the framework default unit crash costs are correctly applied.

2.7.2 Program-Level Validation: Harwood et al. (2003)

The Harwood et al. (2003) study presents a 10-site case study covering a mix of rural and urban undivided and divided nonfreeway facilities (2- and 4-lane), with optimization solved at two budget levels ($10M and $50M). The MCBOMs MILP optimizer was run against the published Harwood case study under the following configuration:

- Harwood's published per-site cost and benefit values from his Tables 2 and 3 (rather than recomputed from raw inputs)

- Harwood's discount rate of 4 percent (per AASHTO 1977)
- Harwood's two-bin severity convention (fatal+injury combined; PDO)
- No deferral-penalty mechanism (PNR is not implemented in the MCBOMs prototype; see the discussion in Section 2.7.2.2 below)

2.7.2.1 $50 Million Budget Case

At the $50M budget level, the budget constraint does not bind: all cost-effective improvements can be funded simultaneously, regardless of any deferral-penalty mechanism. Table 2.3 compares the MCBOMs optimizer's results against the published Harwood values.

Table 2.3: MCBOMs reproduction of Harwood (2003) Table 4, $50 million budget. Differences are attributable to rounding in the published values.

Metric	**Harwood (2003)**	**MCBOMs**	**Difference**
Total benefit (PSB + PTOB)	$10,640,914	$10,640,909	−$5
Total expenditure (resurf + safety)	$16,271,246	$16,271,247	+$1
Safety improvement cost	$4,481,397	$4,481,397	$0
Net benefit	$6,159,517	$6,159,512	−$5
Site selection (alternatives chosen)	10 sites match	10 sites match	—

The MCBOMs optimizer reproduces Harwood's published $50M result to within $5 across all reported metrics, with identical site selection across all 10 sites. The five-dollar discrepancy is consistent with rounding in the intermediate published values. This validates both the project-level MILP formulation (Equations 2.8–2.11) and the safety-benefit calculation (Equation 2.21) at program scale.

2.7.2.2 $10 Million Budget Case and the Treatment of Deferral Penalties

At the $10M budget level, the budget constraint binds: not all sites can be fully funded, and some must be deferred (do-nothing alternative selected). Harwood's published $10M solution defers Sites 4, 6, and 9. The MCBOMs optimizer, configured as described above, instead defers Sites 1, 4, 6, and 8 and produces a higher net benefit. Table 2.4 documents the comparison.

Table 2.4: MCBOMs solution at $10 million budget compared to Harwood (2003) Table 4. The divergence reflects the deferral-penalty methodology described below.

Metric	**Harwood (2003)**	**MCBOMs**	**Difference**
Total benefit (PSB + PTOB)	$7,187,814	$7,600,301	+5.7%
Total expenditure (resurf + safety)	$9,953,579	$9,547,090	−4.1%
Net benefit	$4,675,033	$4,931,520	+5.5%
Sites deferred (do-nothing)	{4, 6, 9}	{1, 4, 6, 8}	differs

This divergence is methodologically intentional and reflects a documented design choice. Harwood's full formulation includes a Penalty for Not Resurfacing (PNR), which adds a deferral cost to the do-nothing alternative roughly proportional to the avoided cost of eventual full pavement reconstruction. The MCBOMs framework does *not* implement PNR for the following reasons:

- Harwood (2003) publishes only the aggregate PNR ($5,576,145 across the deferred sites in the $10M case) and a qualitative formula. Per-site PNR values are not available, so reproducing Harwood's $10M solution exactly would require reverse-engineering site-level penalties rather than computing them from a documented procedure.
- The PNR formula requires per-site pavement condition data and time-to-failure estimates that are not standardized in current FHWA guidance and are not present in the MCBOMs prototype's case study inputs.
- Banihashemi (2007), the second benchmark, also does not use a deferral-penalty mechanism.

Without PNR, the MCBOMs optimizer is free to defer sites strictly on net-benefit grounds, and consequently finds a solution that is 5.5 percent higher in net benefit than Harwood's published $10M solution. Both solutions are correct within their respective formulations: MCBOMs is net-benefit-optimal subject to the budget constraint, while Harwood's solution is optimal subject to budget plus PNR-incentivized resurfacing of additional sites. Highway agencies wishing to incorporate pavement-deferral costs in the optimization can do so by adding those costs to the do-nothing alternative's cost term $C_{i,0}$ as a negative benefit; the MCBOMs formulation supports this directly through Equation 2.8.

2.7.2.3 Cross-Validation Against the Parallel BCA Spreadsheet

Independent of the published-paper validations, the MCBOMs safety benefit output has been cross-validated against the Task 4 data-curation BCA spreadsheet on a per-(segment, alternative) basis. The spreadsheet implements a streamlined formulation:

$$B^{\text{safety, spreadsheet}} = UC_K \cdot (\text{Obs}_K - \text{Proj}_K) + UC_B \cdot (\text{Obs}_{ABC} - \text{Proj}_{ABC}) + UC_O \cdot (\text{Obs}_O - \text{Proj}_O) \tag{2.25}$$

which is Equation 2.18 reduced to (a) a single-year horizon (no discounting), and (b) a three-bin severity classification (K / ABC-combined / O) with the Injury-B unit cost applied to the entire ABC bin. The cross-validation results across 35 segment–alternative pairs in the test dataset are:

- **Replication accuracy.** Maximum absolute error between the MCBOMs Python replication of Equation 2.25 and the spreadsheet's reported *Safety Benefit* values is $0.00. The MCBOMs codebase reproduces every value in the spreadsheet's safety benefit column to within numerical precision.
- **Methodological difference.** The ratio of MCBOMs Equation 2.21 present-value benefit to the spreadsheet's single-year benefit is constant at 10.59 across all 35 pairs, exactly matching $PWF(0.07, 20) = 10.594$. The two tools are mathematically equivalent up to the discounting choice and the severity-binning convention bridge.

2.7.3 Network-Level Validation: Banihashemi (2007)

The Banihashemi (2007) study presents a network case study with three highways (4–10 km in length) and 13 intersections, with the optimization solved at ten budget levels from $1M to $12.2M. The case study uses the IHSDM Crash Prediction Module (CPM) for both highway segment and intersection crash predictions, with delay times computed from the Highway Capacity Manual (Equations A30-11 and A30-15) for intersections and from the published Highway 1 climbing-lane analysis for segments.

Validation scope. The Banihashemi validation is conducted on the intersection sub-problem of his case study: the 13 intersections with published ADTs, improvement combinations, and delay times. The full network reproduction (combining all three highways and all 13 intersections, totaling 135 homogeneous segments and 3,385 binary variables) requires case-study-specific segment geometry data not published in the paper; that full reproduction is documented as a planned extension.

Configuration. The MCBOMs MILP optimizer was run against the Banihashemi intersection sub-problem under the following configuration:

- IHSDM Crash Prediction Module (Equation 15 of Banihashemi), implemented with Banihashemi's published parameters (α, β, γ, C_i) for 3-leg stop, 4-leg stop, and 4-leg signalized intersection types (page 113)
- Intersection type inferred from Table 3 traffic control column: all 13 intersections treated as 4-leg stop-controlled or 4-leg signalized (Int 10) consistent with the traffic control entries
- AMF values for intersection features (skew, traffic control change, left-turn lane, sight distance deficiency) from IHSDM and Vogt/Bared base models; note that Banihashemi does not publish his exact AMF values, so exact numerical reproduction is not expected
- Unit delay cost $12.50 per vehicle-hour (Banihashemi page 109)
- Average crash costs by facility type: $55,239 (3-leg stop), $81,375 (4-leg stop), $31,665 (4-leg signalized) (Banihashemi page 109)
- 20-year project lifetime, no discounting (per Banihashemi's framework)

Results. Table 2.5 shows the MCBOMs intersection selections at each budget level studied by Banihashemi.

Table 2.5: MCBOMs intersection sub-problem results compared to the intersection portion of Banihashemi (2007) Table 5. Notation: "Int x: LTL" means alternative with left-turn lane added at intersection x; "SA" means skew-angle correction; "ISD" means sight distance improvement.

Budget	MCBOMs (intersection sub-problem)	Banihashemi (full network)
Unlimited	Int 1:LTL; Int 2:LTL; Int 5:LTL; Int 6:LTL; Int 7:SA,ISD; Int 9:LTL; Int 12:LTL	Int 1:LTL; Int 2:LTL; Int 5:LTL; Int 6:LTL; Int 7:SA,ISD; Int 9:LTL; Int 10:LTL; Int 12:SA,LTL; Int 13:SA,LTL,ISD
$8M	same as unlimited (spent $3.2M)	Int 1:LTL; Int 2:LTL; Int 5:LTL; Int 6:LTL; Int 12:LTL; Int 13:LTL
$6M	same as unlimited (spent $3.2M)	Int 7:SA,ISD; Int 12:LTL; Int 13:LTL
$3M	Int 1:LTL; Int 2:LTL; Int 5:LTL; Int 6:LTL; Int 9:LTL; Int 12:LTL (spent $2.1M)	Int 12:LTL; Int 13:LTL
$1M	Int 1:LTL; Int 2:LTL; Int 12:LTL (spent $0.7M)	Int 12:LTL (spent within $1M for intersections only)

Interpretation of the comparison. The MCBOMs optimizer and Banihashemi's published solutions agree on three structural findings:

1. **The most cost-effective improvement is Int 12:LTL.** Both the MCBOMs optimizer and Banihashemi identify the left-turn lane at Intersection 12 as the highest-benefit-per-dollar improvement (benefit-to-cost ratio of approximately 11.5 in the MCBOMs computation; consistently selected at every non-zero budget in Banihashemi Table 5).

2. **Signalization of Intersections 3 and 4 is never selected.** Both solvers correctly identify that signalizing these moderate-traffic minor-stop intersections would increase delay substantially (from 6,078 to 98,165 and 72,450 vehicle-hours/year respectively), making the delay cost increase dominate the crash cost reduction. Neither Banihashemi's solution nor MCBOMs selects these alternatives at any budget level.

3. **Rank ordering of LTL improvements matches.** In both solutions, Intersections 12, 1, 2, 5, 6 appear as the first cost-effective LTL candidates as budget expands.

The two solutions *differ* in two respects that are attributable to the scope of the sub-problem and to information not published in the original paper:

1. **Budget coupling with segment improvements.** Banihashemi's Table 5 reports results for the full network; his intersection selections reflect the fact that segment improvements (lane widening, shoulder widening, etc.) are competing for the same budget. When isolated from the segment problem, the MCBOMs optimizer has the full budget available for intersections and selects more of them at any given budget level. This is a structural consequence of validating against a sub-problem rather than the full network.

2. **AMF values not published.** Banihashemi does not publish the specific AMF values used for LTL, traffic control changes, or sight distance deficiency. The MCBOMs reproduction uses standard IHSDM and Vogt/Bared AMF values, which may differ from Banihashemi's choices by 5–15 percent. This difference can shift the optimal alternative selection at the margin (for example, Intersection 10 LTL and several higher-cost alternatives at Intersections 12 and 13 that Banihashemi selects are marginal in the MCBOMs reproduction). Exact numerical reproduction of Banihashemi's selections at all budget levels would require his specific AMF values, which are not available.

Validation conclusion. The MCBOMs MILP framework correctly implements the optimization structure used by Banihashemi (2007), as evidenced by (a) consistent identification of the most cost-effective improvement (Int 12:LTL) as the first-selected alternative in both solutions, (b) consistent rejection of signalization at Intersections 3 and 4 on delay-cost grounds, and (c) consistent rank ordering of LTL improvements across the intersection set. Residual divergence between the two solutions is attributable to the intersection-sub-problem scope and to Banihashemi's unpublished AMF values, neither of which constitutes a defect in the MCBOMs formulation or implementation.

2.7.4 Summary of Validation Evidence

The MCBOMs framework has been validated at three independent levels:

1. **Methodology level** (Section 2.7.1): direct algebraic reproduction of the worked safety-benefit example. Match within floating-point precision.

2. **Program level** (Section 2.7.2): full optimization reproduction of the Harwood (2003) $50M published case to within $5 (rounding) on all metrics, with identical site selection across all 10 sites. At the $10M budget, the MCBOMs optimizer produces a strictly net-benefit-superior solution because the framework does not implement Harwood's PNR deferral-penalty mechanism; this methodological choice is documented and defensible.

3. **Network level** (Section 2.7.3): reproduction of the Banihashemi (2007) intersection sub-problem. The MCBOMs MILP correctly identifies the most cost-effective intersection improvement (Int 12:LTL) as the first-selected alternative, correctly rejects signalization at Intersections 3 and 4 on delay-cost grounds, and produces a rank ordering of LTL improvements consistent with Banihashemi's published Table 5. Residual numerical divergence is attributable to the intersection-sub-problem scope (versus Banihashemi's full-network solution) and to AMF values not published in the original paper.

Cross-validation. In addition to the published-paper benchmarks, per-segment cross-validation against the Task 4 data-curation BCA spreadsheet shows perfect replication ($0.00 maximum error across 35 segment–alternative pairs) of the spreadsheet's safety benefit calculation, with the constant 10.59× ratio of present-value to single-year benefit confirming methodological equivalence up to discounting.

Reproducibility. All validation procedures described in this document are reproducible from the project's GitHub repository. The automated test suite includes 109 tests covering the worked example, the Harwood reproduction, and unit tests for each safety-benefit computational component. The Harwood validation script (`run_harwood_validation.py`) prints a complete validation summary on each run, including informational reporting of the documented $10M divergence.